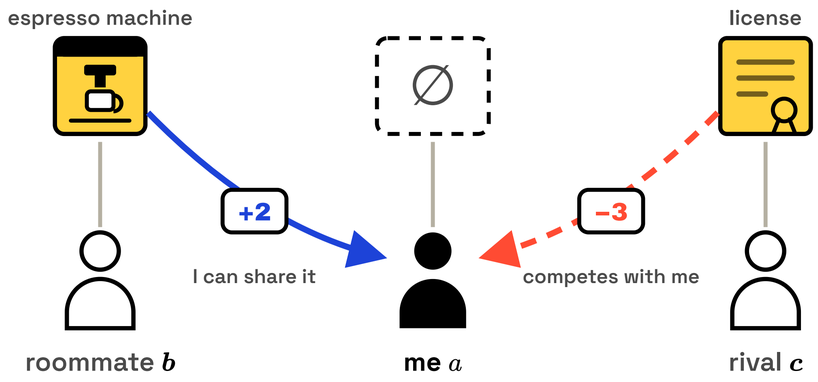
Tight Asymptotic Bounds for
Fair Division with Externalities

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1 THE MODEL — AN EXAMPLE



$$V_a(A) = (+2) + (-3) = -1$$

I receive **nothing** — yet the allocation matters to me

$$V_a(A) = \sum_{x \in M} V_a\big(A(x),\, x\big)$$

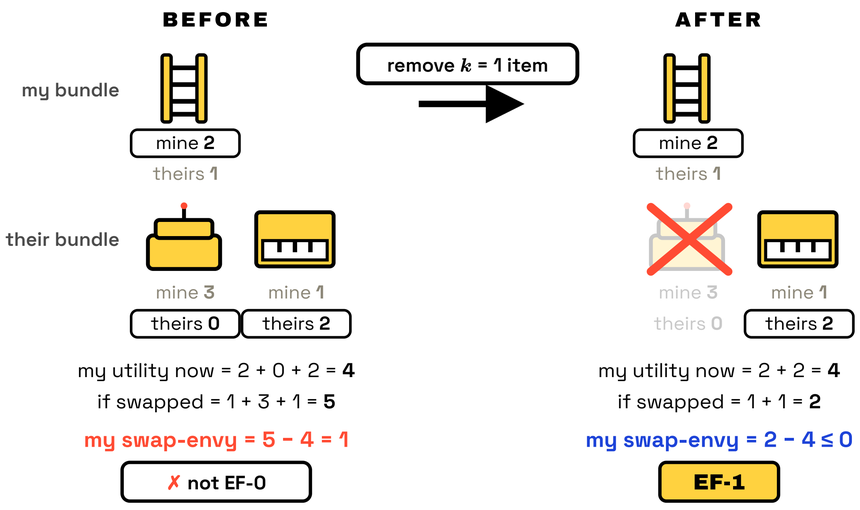
$A(x)$ — the agent holding item x

indivisible items · additive valuations · goods & chores

2 THE NOTION — EF-k

EF-k : for every pair of agents, removing $\le k$ items eliminates swap-envy.

all numbers are **my** utilities — **mine** / **theirs** says who holds the item



smaller k = fairer · EF-0 = fully envy-free · one yardstick for any allocation

3 THE QUESTION

Does an EF1 allocation always exist?

STATE OF THE ART — WITH EXTERNALITIES

	EF	EF1	optimal k
no externalities BASELINE · ANY n	✗	✓ Lipton et al. '04	$k = 1$
n = 2	✗	✓ Aziz et al. '23	$k = 1$
n = 3	✗	✓ binary · no chores — Aziz et al. '23 ? otherwise — open	$k = 1$ binary · no chores
general n THIS PAPER	✗	?	?

EF can fail already for two agents and one good · Deligkas et al. '24 conjectured EF1 for binary valuations.

4 THE ANSWER

EF-$\Theta(\sqrt{n})$

the optimal envy relaxation, over all instances with n agents

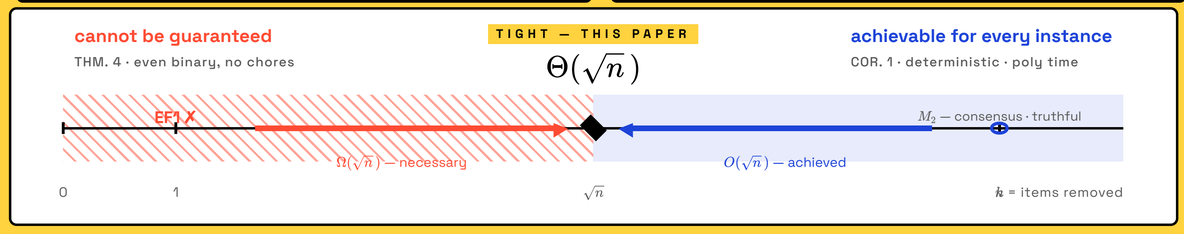
✗ EF1 CAN FAIL
refutes the Deligkas et al. '24 conjecture · settles Aziz et al. '23

ALWAYS ACHIEVABLE · COR. 1

EF-$O(\sqrt{n})$ exists for **every** instance — deterministic, polynomial time.

UNAVOIDABLE · THM. 4 + 6

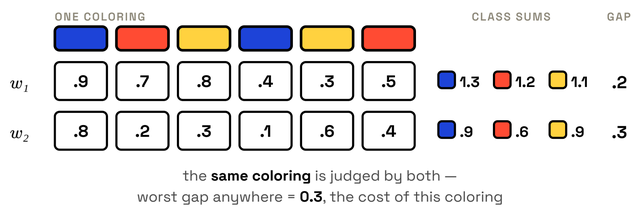
$\Omega(\sqrt{n})$ is necessary — even **binary, no chores**, all externalities aimed at a **single agent**.



5 MULTICOLOR DISCREPANCY

what it is

color a ground set of m elements with q colors: $\chi : [m] \to [q]$. a **single** coloring is judged by **every** weight vector $w \in [0,1]^m$: each pair of color classes must carry almost equal weight. $\mathrm{wdisc}_{q,\infty}$ is the best achievable worst-case gap.



the **same coloring** is judged by both —
worst gap anywhere = **0.3**, the cost of this coloring

the dictionary

ADDITIVE FAIR DIVISION — NO EXTERNALITIES	⇄	DISCREPANCY
agents	⇄	colors
items	⇄	elements
valuations	⇄	weight vectors
allocation	⇄	q-coloring
swap-envy of a pair	⇄	imbalance of a color pair

with externalities this direct map **breaks** — **Prop. 1** restores it: each ordered pair's asymmetric envy values v_{ij} become the weight vectors:

THE BRIDGE · PROP. 1 $\quad V_i\big(A^{i \leftrightarrow j}\big) - V_i(A) = v_{ij}(A_j) - v_{ij}(A_i)$

swap-envy with externalities ⇄ pairwise bundle envy — bounds transfer both ways

ACHIEVABLE · LEVY ET AL. '17 + MANURANGSI–MEKA '26

a coloring with pairwise imbalance $O(\sqrt{q})$ always exists — **derandomized**, polynomial time.

TIGHT · EXTENDS MANURANGSI–MEKA '26

$\mathrm{wdisc}_{q,\infty} \ge \frac{\sqrt{q}-1}{16}$ — no coloring does better.

with $q = n$ colors this is exactly $\Theta(\sqrt{n})$.

6 UPPER BOUND

• **one coloring balances every pair**

1 EXTERNALITIES
$V_i(j, x)$

→

2 PROP. 1 · EQUIVALENCE
asymmetric envy $v_{ij}(x) = V_i(i, x) - V_i(j, x)$

↓

3 LEMMA 1
4 valuations in $[0, 1]$ — goods & chores together

←

4 MANURANGSI–MEKA '26 · LEVY ET AL. '17
multicolor discrepancy coloring, derandomized

↓

5 COR. 1
EF-$O(\sqrt{n})$ · deterministic · poly time

7 LOWER BOUND

• **from discrepancy back to envy**

1 HARD INSTANCE
Hadamard blocks $\frac{1}{2}(\mathbf{1}+H)$ — binary valuations

↓

2 MANURANGSI–MEKA '26
$\mathrm{wdisc}_{1/n} \ge \frac{\sqrt{q}-1}{16}$

→

3 LEMMA 2
EF-c $\Rightarrow$ $\mathrm{wdisc} \le 6c$

↓

4 THM. 4 · 5 · 6
$c = \Omega(\sqrt{n})$ · binary · no chores · one-target

8 TWO MECHANISMS

	M_1 DETERMINISTIC	M_2 RANDOMIZED
fairness	EF-$O(\sqrt{n})$	EF-$O(n)$
consensus bundles	—	✓ permute freely
truthful	—	✓ in expectation
runtime	polynomial	polynomial

M_2 assigns bundles uniformly at random — no agent (or coalition) can shift its expected value by misreporting. Extends Bu–Tao FOCS '25 to externalities.

9 OPEN

? a small **explicit** counterexample to EF1
what is the largest n where EF1 always exists?

? a matching $\Omega(n)$ **consensus** lower bound
is M_2 asymptotically optimal?

? other **fairness** notions
do EFX, proportionality, or MMS admit similar tight bounds with externalities?

? beyond externalities
does the discrepancy toolbox transfer to social-impact fair division?

10 REFERENCES

R. J. Lipton, E. Markakis, E. Mossel, A. Saberi. **On approximately fair allocations of indivisible goods.** EC '04.

A. Deligkas, E. Eiben, V. Korchemna, Š. Schierreich. **The complexity of fair division of indivisible items with externalities.** AAAI '24.

A. Levy, H. Ramadas, T. Rothvoss. **Deterministic discrepancy minimization via the multiplicative weight update method.** IPCO '17.

H. Aziz, W. Suksompong, Z. Sun, T. Walsh. **Fairness concepts for indivisible items with externalities.** AAAI '23.

P. Manurangsi, R. Meka. **Tight lower bound for multicolor discrepancy.** SOSA '26.

X. Bu, B. Tao. **Truthful and almost envy-free mechanism of allocating indivisible goods: the power of randomness.** FOCS '25.

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PAPER
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